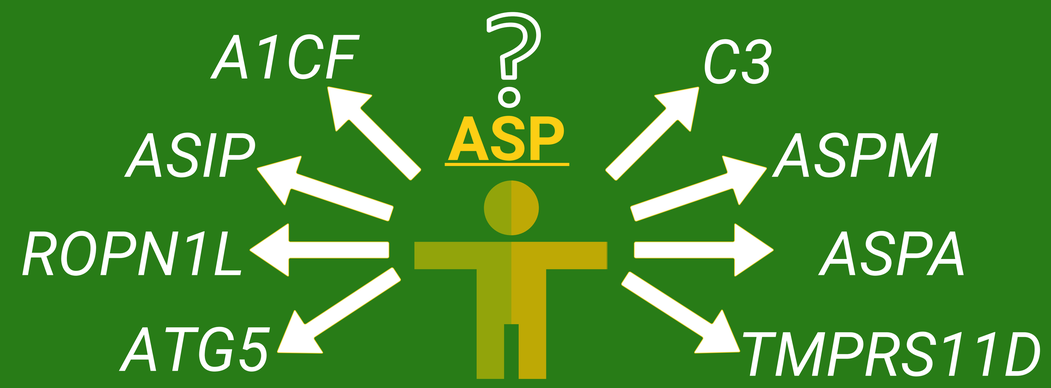
Gene nomenclature aids in **clear** and **consistent** communication about genomic knowledge.



When it creates **confusion**, how do we resolve it?

Curating Gene Alias Relationships to Resolve Human Gene Symbol Ambiguity

Nationwide Children's — When your child needs a hospital, everything matters.

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The Ohio State University — Wexner Medical Center

NIH R00 HG010157

Alias-Alias Collisions occur between genes that share an alias symbol

HGNC ID	Gene Symbol	Aliases
19048	ASPM	MCPH5, ASP, Calmbp1, FLJ10517, FLJ10549
1318	C3	C3a, ASP, CPAMD1, ARMD9, C3b
24059	TMPRSS11D	HAT, ASP

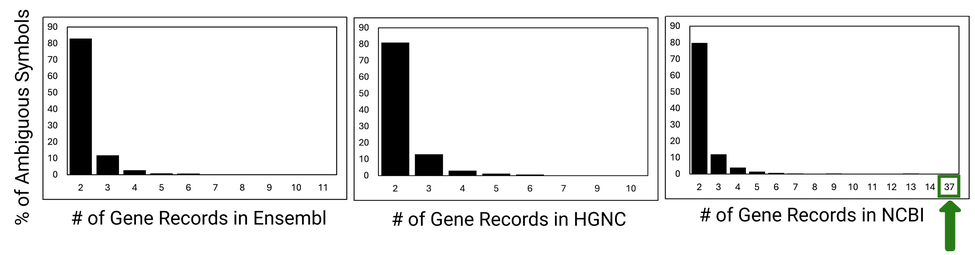
Alias-Primary Collisions occur between genes that share an alias and a primary symbol

HGNC ID	Gene Symbol	Aliases
7989	NRAS	NS6, CMNS, KRAS, N-ras, NCMS, NRAS1, ALPS4
6407	KRAS	NS, NS3, OES, CFC2, RALD, K-Ras, KRAS1...

Collisions occur in Ensembl, HGNC, and NCBI databases

Source	# of gene records	% of gene records with alias-alias collisions	% of gene records with alias-primary collisions
Ensembl	41,068	6.16% (2,530)	1.65% (678)
HGNC	45,646	6.82% (3,113)	1.40% (637)
NCBI	45,390	13.25% (6,013)	3.72% (1,689)

Ambiguous symbol distribution in different databases

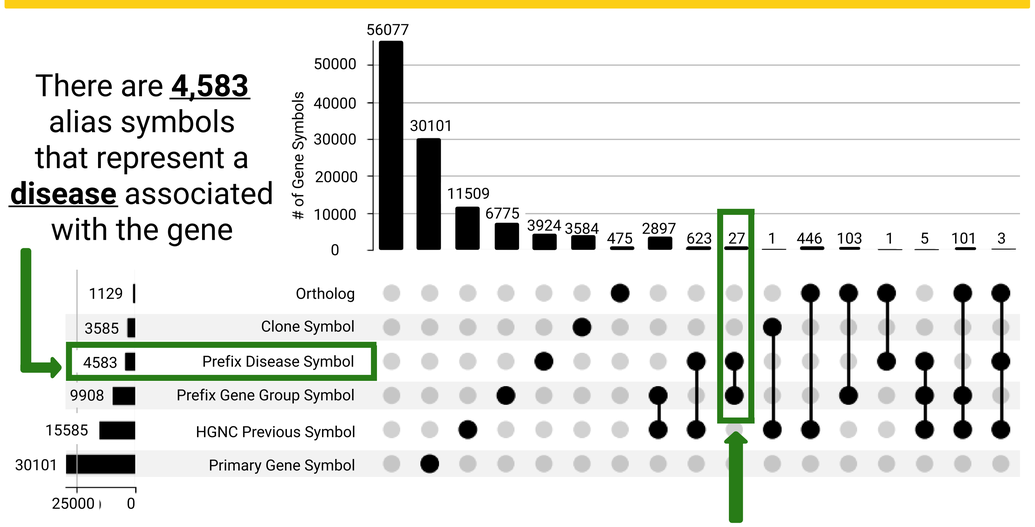


Most ambiguous symbols are shared between **2** gene records

In NCBI, *VH* is shared between **37** gene records

We have annotated 52% of all Gene-Symbol relationships

There are **4,583** alias symbols that represent a **disease** associated with the gene



There are **27** alias symbols that represent a **disease** the gene is associated with & the **gene group** of the gene

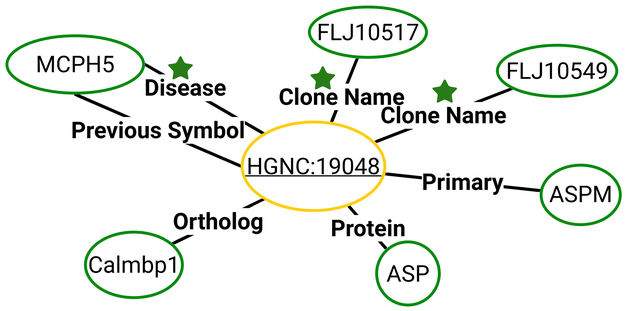
Gene-Symbol Relationship Examples

A gene record has a **list of aliases** with **no information** about the symbols' relationship **to the gene**

Symbol Report for ASPM
Primary gene symbol: ASPM
Gene name: assembly factor for spindle microtubules
HGNC ID: 19048
Chromosomal location: 1q31.3
Alias Symbols: MCPH5, ASP, Calmbp1, FLJ10517, FLJ10549

Resources outside of the gene record can identify the **relationship** between the **alias symbol** and the **gene**

Disease Database
OMIM # 608716
MICROCEPHALY 5, PRIMARY, AUTOSOMAL RECESSIVE: MCPH5
Phenotype-Gene Relationships:
Location- 1q31.3
Gene/Locus: ASPM

Clone Database
FLJ Human cDNA Database
FLJ25096 FLJ10549
FLJ20472 FLJ10517
FLJ43933 FLJ91001
FLJ43971 FLJ91002
FLJ20587 FLJ91003

After annotation, the **relationships** (★) are now defined between the **symbols** and the **genes** and can be used to **resolve ambiguities**